THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Carbon and its Compounds** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 4	O=C=O has two double bonds: 4 shared pairs.
2	**(a)** long-chain carboxylic acids	Fatty acid salts.
3	**(a)** catenation	Catenation.
4	**(d)** 16	3n + 1 = 16.
5	**(d)** formation of micelles	Hydrophobic tails trap oil in micelles.
6	**(a)** four valence electrons	Difficult to gain or lose 4 electrons.
7	**(d)** C_2H_4	Ethene has a double bond.
8	**(c)** calcium and magnesium salts forming scum	Insoluble Ca/Mg salts of fatty acids.
9	**(a)** 2	n-butane and isobutane.
10	**(d)** C_4H_8	Homologues differ by CH_2.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Carbon has four valence electrons (1); Cannot easily lose or gain 4 electrons, so it shares them (1).	1+1
12	Atom or group that decides chemical properties (1); -OH (alcohol) (1).	1+1
13	Acid reacts with alcohol to give ester and water (1); $CH_3COOH + C_2H_5OH \longrightarrow CH_3COOC_2H_5 + H_2O$ (1).	1+1
14	Self-linking of atoms to form chains/rings (1); Carbon-carbon bonds are very strong (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	n-butane $CH_3CH_2CH_2CH_3$ (1); isobutane (2-methylpropane) $CH_3CH(CH_3)CH_3$ (1); Both are C_4H_{10} (1).	1+1+1
16	Ethanol is oxidised by alkaline $KMnO_4$ or acidified $K_2Cr_2O_7$ (1); $CH_3CH_2OH \longrightarrow CH_3COOH$ (1); Oxidising agent adds oxygen (1).	1+1+1
17	Saturated: single bonds only; C_2H_6 (1); Unsaturated: double or triple bonds; C_2H_4 (1); Unsaturated show addition reactions (1).	1+1+1
	OR: C_2H_4, $CH_2{=}CH_2$ (1); Contains a carbon-carbon double bond (1); Can add hydrogen or halogen (1).	1+1+1
18	$CH_3COOH + NaOH \longrightarrow CH_3COONa + H_2O$ (1); $2\,CH_3COOH + Na_2CO_3 \longrightarrow 2\,CH_3COONa + H_2O + CO_2$ (2).	1+2
	OR: Diamond: each carbon bonded to four others in 3D (1); Graphite: layers of hexagons with weak forces between (1); Layers slide over each other (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Carbon has 4 valence electrons (1); Forms covalent bonds by sharing (1); Catenation forms chains and rings (1); Tetravalency: bonds with 4 atoms (1); CH_4: C shares 4 pairs with 4 H (1).	1+1+1+1+1
	OR: $CH_3COOH + C_2H_5OH \longrightarrow CH_3COOC_2H_5 + H_2O$ (1); Conc. H_2SO_4 catalyst (1); Esters have fruity smell (1); Used in perfumes and flavours (1); $CH_3COOC_2H_5 + NaOH \longrightarrow CH_3COONa + C_2H_5OH$ (1).	1+1+1+1+1
20	Same molecular formula C_5H_{12} (1); n-pentane (1); 2-methylbutane (isopentane) (1); 2,2-dimethylpropane (neopentane) (1); Three isomers (1).	1+1+1+1+1
	OR: Hydrophilic head, hydrophobic tail (1); Tail dissolves in oil (1); Micelle formation (1); Hard water gives scum (1); Detergents avoid scum (1).	1+1+1+1+1

Mistakes to avoid:

- Counting covalent bonds wrongly: an alkane C_nH_{2n+2} has $3n + 1$ bonds (include all C-H and C-C bonds).
- Confusing ethanol and ethanoic acid: ethanol (C_2H_5OH) is an alcohol, ethanoic acid (CH_3COOH) releases CO_2 with carbonates; ethanol does not.
- Naming the chain from the wrong carbon count, e.g. calling CH_3CH_2CHO 'propanol' instead of propanal; the suffix comes from the functional group (-ol, -al, -one, -oic acid).

- Writing an addition reaction for a saturated hydrocarbon; saturated compounds undergo substitution, unsaturated ones undergo addition.
- Saying that detergents form micelles but soaps do not, or that soap works well in hard water; soaps give scum in hard water.
- Stating that diamond and graphite conduct electricity; only graphite does (it has free electrons).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Carbon and its Compounds** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** an alcohol	Esterification.
2	**(d)** non-conductor and very hard	Each carbon bonded to four others.
3	**(d)** aldehyde	-CHO is the aldehyde group.
4	**(c)** sweet fruity smell	Used in perfumes.
5	**(b)** 60	C_{60}.
6	**(c)** ethanol	Two carbons plus -ol.
7	**(c)** sodium hydroxide	Gives alcohol and soap.
8	**(c)** its layers slide over each other	Weak forces between layers.
9	**(a)** carbon dioxide	Acids liberate CO_2 from carbonates.
10	**(d)** nickel	Ni catalyst converts oils to fats.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Series of compounds with the same functional group differing by CH_2 (1); Alkanes: CH_4, C_2H_6, C_3H_8 (1).	1+1
12	Alkaline hydrolysis of an ester (1); Gives soap and alcohol (1).	1+1
13	CH_3-CH_3, each C shares 4 pairs (1); 7 covalent bonds (6 C-H and 1 C-C) (1).	1+1
14	Needs to lose or gain 4 electrons which requires too much energy (1); Sharing is easier (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Chloromethane (1); Propanal (1); Propanoic acid (1).	1+1+1
16	Unsaturated have a higher percentage of carbon (1); Incomplete combustion gives soot (1); Saturated burn completely (1).	1+1+1
17	Butane C_4H_{10} (1); Pentane C_5H_{12} (1); Difference CH_2 (1).	1+1+1
	OR: Add sodium hydrogencarbonate (1); Ethanoic acid gives brisk effervescence of CO_2 (1); Ethanol gives none (1).	1+1+1
18	Diamond: jewellery, cutting (1); Graphite: lubricant, electrodes (1); Fullerene: nanotechnology (1).	1+1+1
	OR: They are covalent compounds (1); No ions are present (1); Weak intermolecular forces (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Ionic: high melting point, e.g. NaCl (1); Covalent: low, e.g. CH_4 (1); Ionic conduct in solution (1); Covalent do not (1); Solubility: ionic in water, covalent in organic solvents (1).	1+1+1+1+1
	OR: Halo: -Cl, CH_3Cl (1); Alcohol: -OH, CH_3OH (1); Aldehyde: -CHO, CH_3CHO (1); Ketone: C=O, CH_3COCH_3 (1); Acid: -COOH, CH_3COOH (1).	1+1+1+1+1
20	$CH_4 + 2\,O_2 \longrightarrow CO_2 + 2\,H_2O$ (1); $CH_3CH_2OH \longrightarrow CH_3COOH$ (1); $CH_2{=}CH_2 + H_2 \longrightarrow CH_3CH_3$ (1); $CH_4 + Cl_2 \longrightarrow CH_3Cl + HCl$ (1); Addition for unsaturated; substitution for saturated (1).	1+1+1+1+1
	OR: Methane CH_4 (1); Ethane C_2H_6 (1); Propane C_3H_8 (1); Butane C_4H_{10} (1); General formula $CnH_2n_2^+$ (1).	1+1+1+1+1

Mistakes to avoid:

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- Naming the chain from the wrong carbon count, e.g. calling CH_3CH_2CHO 'propanol' instead of propanal; the suffix comes from the functional group (-ol, -al, -one, -oic acid).
- Writing an addition reaction for a saturated hydrocarbon; saturated compounds undergo substitution, unsaturated ones undergo addition.
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ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Carbon and its Compounds** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** hydrogen	$2\,Na + 2\,C_2H_5OH \longrightarrow 2\,C_2H_5ONa + H_2$.
2	**(c)** ethanoic acid	5-8% acetic acid.
3	**(d)** unsaturated hydrocarbons	Double/triple bonds open up.
4	**(d)** CH_3COCH_3	Propanone.
5	**(c)** ethene	Dehydration of ethanol.
6	**(a)** saturated hydrocarbons	Alkanes react with chlorine in sunlight.
7	**(d)** CH_4, C_2H_6, C_3H_8	Same general formula, differ by CH_2.
8	**(b)** C_2H_2	Unsaturated compounds have more carbon percentage.
9	**(d)** ethanoic acid	Oxidation of alcohol to acid.
10	**(d)** CnH_2n-2	Alkynes have a triple bond.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Ca and Mg salts of hard water react with soap (1); Insoluble precipitate called scum forms (1).	1+1
12	Compounds with same molecular formula but different structures (1); Butane and isobutane (C_4H_{10}) (1).	1+1
13	CO_2 and water (1); $2\,C_2H_6 + 7\,O_2 \longrightarrow 4\,CO_2 + 6\,H_2O$ (1).	1+1
14	Detergents are sodium salts of sulphonic acids (1); Their Ca and Mg salts are soluble, so no scum (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Soap molecule has hydrophilic head and hydrophobic tail (1); Tails attach to oil, heads in water, forming micelles (1); Micelles carry dirt away on rinsing (1).	1+1+1
16	Addition of hydrogen to unsaturated compounds with nickel catalyst (1); $CH_2{=}CH_2 + H_2 \longrightarrow CH_3CH_3$ (1); Used to make vanaspati ghee from oils (1).	1+1+1
17	A hydrogen atom is replaced by another atom (1); $CH_4 + Cl_2 \longrightarrow CH_3Cl + HCl$ (1); Occurs in sunlight (1).	1+1+1
	OR: Sodium ethoxide and hydrogen (1); $2\,C_2H_5OH + 2\,Na \longrightarrow 2\,C_2H_5ONa + H_2$ (2).	1+2
18	Ethanol burns with a clean blue flame (1); $C_2H_5OH + 3\,O_2 \longrightarrow 2\,CO_2 + 3\,H_2O$ (1); Releases heat and light (1).	1+1+1
	OR: Oxidation of ethanol (1); Pure ethanoic acid freezes at 290 K (1); So it is called glacial acetic acid (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Colourless liquid, soluble in water (1); $2\,C_2H_5OH + 2\,Na \longrightarrow 2\,C_2H_5ONa + H_2$ (1); Dehydration at 443 K with conc. H_2SO_4 gives ethene (1); Uses: solvent, fuel, medicines (1); Harmful effects of drinking (1).	1+1+1+1+1
	OR: Diamond: 3D network, hardest, insulator (1); Graphite: layers, conducts, soft (1); Fullerene: C_{60}, football shape (1); All made only of carbon (1); Uses of each (1).	1+1+1+1+1
20	$CH_3COOH + NaOH \longrightarrow CH_3COONa + H_2O$ (1); With carbonate gives CO_2 (1); Esterification with ethanol (1); Vinegar is 5-8% (1); Uses: food preservation, vinegar (1).	1+1+1+1+1
	OR: Saturated burn with clean flame (1); Unsaturated burn with sooty flame (1); Bromine water is decolourised by unsaturated compounds (1); Addition reaction takes place (1); Saturated do not decolourise bromine water (1).	1+1+1+1+1

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